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PSAC Exam Practice · Predicted Specimen Paper

2026

SCIENCE

Grade 5 · Paper 1

Food Chains | States of Matter | Light and Shadow

Time Allowed: **1 hour**

Total Marks: **50**

Name: _____

Class: _____ Date: _____

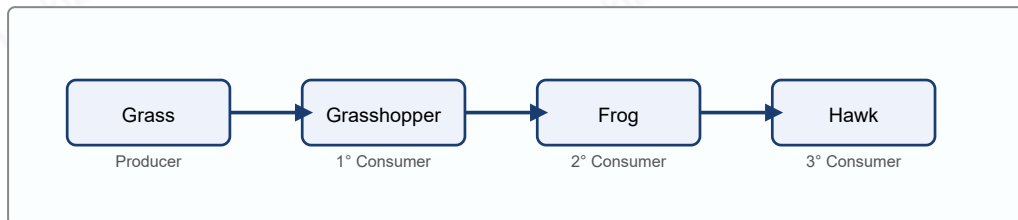
INSTRUCTIONS TO CANDIDATES

1. Write your name, class and date in the spaces provided above.
2. Answer **ALL** questions in **all three sections**.
3. Write your answers clearly in the spaces provided.
4. The mark for each question or part question is shown in square brackets [].
5. Diagrams should be clear and neatly labelled.

Section	Topic	Available	Score
A	Living Things — Food Chains	20	
B	Matter and Materials — States of Matter	15	
C	Energy and Forces — Light and Shadow	15	
TOTAL		50	

SECTION A — Living Things: Food Chains [20 marks]

- 1 Study the food chain diagram below and answer the questions that follow.



- (a) Which organism is the *producer* in this food chain? [1]

- (b) Which organism is the *secondary consumer*? [1]

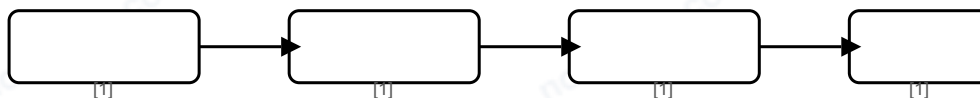
- (c) What does the arrow in a food chain represent? [1]

- (d) Which organism is the *top predator*? [1]

- (e) Name the source of energy for *all* food chains. [1]

- 2 Use the words in the box to complete the food chain. Write one word in each box.

eagle grass rabbit snake



[Total: 4 marks]

- 3 Tick **ONE** box to show the correct answer for each question. [4]

(a) A plant is called a _____ because it can make its own food using sunlight.

- ☐ consumer
- ☐ producer
- ☐ decomposer
- ☐ predator

(b) An animal that eats ONLY plants is called a:

- ☐ carnivore
- ☐ herbivore
- ☐ omnivore
- ☐ scavenger

(c) Decomposers are important because they:

- ☐ produce food from sunlight
- ☐ break down dead organisms and return nutrients to the soil
- ☐ eat large predators
- ☐ carry energy to plants

(d) In the food chain: **leaves** → **caterpillar** → **robin** → **fox**, the robin is a:

- ☐ producer
- ☐ primary consumer
- ☐ secondary consumer
- ☐ tertiary consumer

4 Write **TRUE** or **FALSE** for each statement.

[4]

Statement	True or False
All food chains begin with a plant (producer).	
Carnivores eat only plants.	
A food web shows how several food chains are linked together.	
Energy is passed from prey to predator along a food chain.	

5 In a grassland, the food chain is: **grass** → **zebra** → **lion**.

(a) A disease kills most of the zebras. Explain what would happen to the *lion* population.

[1]

- (b) The grass population grows very large after the zebras disappear. Suggest **one** reason for this.

[1]

- (c) Name **one** organism that could be added to this food chain as a decomposer.

[1]

SECTION B — Matter and Materials: States of Matter [15 marks]

6 Answer these questions about the three states of matter.

[4]

(a) Which state of matter has a *fixed shape* and a *fixed volume*?

(b) Which state of matter takes the *shape of its container* but has a fixed volume?

(c) Which state of matter has *no fixed shape* and *no fixed volume*, and fills its container completely?

(d) Give **one** example of a gas found in the air.

7 Write the correct state of matter — **solid**, **liquid** or **gas** — for each example.

[4]

Example	State of Matter
A wooden block	
Orange juice in a glass	
Steam rising from a kettle	
An ice cube	

8 Tick **ONE** box to show the correct answer.

[4]

(a) When water is heated to 100°C, it changes to:

☐ ice ☐ steam ☐ snow ☐ oil

(b) The change from water vapour back to liquid water is called:

☐ melting ☐ evaporation ☐ condensation ☐ freezing

(c) The temperature at which ice melts is:

☐ 0°C ☐ 10°C ☐ 100°C ☐ 37°C

(d) Which of the following is a reversible change of state?

☐ Burning paper ☐ Cooking an egg ☐ Melting ice

☐ Burning paper ☐ Cooking an egg ☐ Melting ice

☐ Rusting iron

9 A student placed an ice cube on a plate in a sunny room.

(a) Name the change of state that happens when the ice cube turns into water. [1]

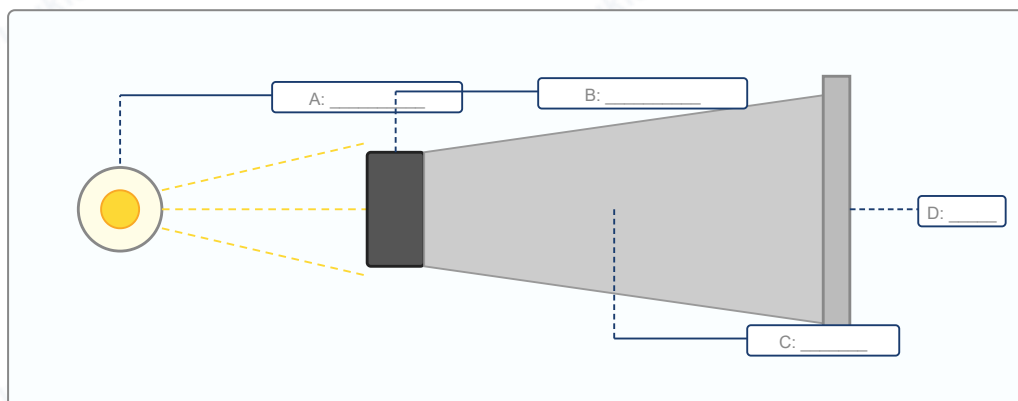
(b) Name the change of state that would happen if the water was heated until it became steam. [1]

(c) Is this change *reversible* or *irreversible*? Give a reason for your answer. [1]

SECTION C — Energy and Forces: Light and Shadow [15 marks]

- 10 The diagram below shows how a shadow is formed. Use the words in the box to label the diagram. [4]

light source opaque object shadow screen



Label A, B, C and D using the words in the box. [1 mark each]

- 11 Tick **True** or **False** for each statement about light and shadows. [4]

Statement	True	False
Light travels in straight lines.	☐	☐
A shadow is formed when light passes through a transparent object.	☐	☐
Moving the object closer to the light source makes the shadow larger.	☐	☐
An opaque object allows all light to pass through it.	☐	☐

- 12 Answer these questions about light and shadow. [4]

(a) Why does a shadow have a similar shape to the object blocking the light? [1]

(b) What happens to the size of the shadow when the object is moved *further* from the light source? [1]

(c) Give **one** example of a *transparent* material.

[1]

(d) Give **one** example of a *translucent* material.

[1]

13 A student shines a torch at a white wall in a dark room. She then places a *thick book* between the torch and the wall.

(a) What will appear on the wall?

[1]

(b) Explain why this happens.

[1]

(c) Name the type of material (transparent, translucent or opaque) that the book is made of.

[1]

END OF PAPER 1 · Total Marks: 50

MARK SCHEME — Grade 5 Science · Paper 1

For teacher / parent use only. Accept any reasonable scientific answer that conveys the same meaning. Award full marks where a candidate's response is scientifically correct even if worded differently from the model answer. Do not penalise poor spelling unless it changes the scientific meaning.

SECTION A — Food Chains [20 marks]

Q	Part	Acceptable Answer	Marks
1	(a)	Grass	1
	(b)	Frog	1
	(c)	The arrow shows the direction of energy/food transfer (from prey to predator / from eaten to eater)	1
	(d)	Hawk	1
	(e)	The Sun	1
2	—	grass → rabbit → snake → eagle (1 mark per correct organism in correct position)	4
3	(a)	producer	1
	(b)	herbivore	1
	(c)	break down dead organisms and return nutrients to the soil	1
	(d)	secondary consumer	1
4	—	TRUE / FALSE / TRUE / TRUE	1 each (4 total)
5	(a)	The lion population would decrease / lions would starve because their food source (zebra) has disappeared.	1
	(b)	There are no/fewer zebras to eat the grass, so grass grows unchecked.	1
	(c)	Any named decomposer: fungi / mushrooms / bacteria / worms (accept earthworms)	1

SECTION B — States of Matter [15 marks]

Q	Part	Acceptable Answer	Marks
6	(a)	Solid	1
	(b)	Liquid	1
	(c)	Gas	1

Q	Part	Acceptable Answer	Marks
	(d)	Any correct gas: oxygen / nitrogen / carbon dioxide / helium / hydrogen	1
7	—	Solid / Liquid / Gas / Solid (1 mark each)	4
8	(a)	steam	1
	(b)	condensation	1
	(c)	0°C	1
	(d)	Melting ice	1
9	(a)	Melting	1
	(b)	Evaporation / boiling	1
	(c)	Reversible — because the steam can be cooled back to water / water can be frozen back to ice.	1

SECTION C — Light and Shadow [15 marks]

Q	Part	Acceptable Answer	Marks
10	A–D	A: light source · B: opaque object · C: shadow · D: screen (1 mark each)	4
11	—	TRUE / FALSE / TRUE / FALSE (1 mark each)	4
12	(a)	Light travels in straight lines, so it is blocked in the same outline/shape as the object.	1
	(b)	The shadow gets smaller / decreases in size.	1
	(c)	Any correct example: glass / clear plastic / water / air	1
	(d)	Any correct example: frosted glass / tissue paper / wax paper / some curtains	1
13	(a)	A shadow (in the shape of the book)	1
	(b)	The book blocks the light / light cannot pass through the book, so a shadow forms on the wall.	1
	(c)	Opaque	1